Callie Clark

callieclark@nyu.edu | (831) 227-9754 | linkedin.com/in/callie-clark/ | github.com/callieclark | Brooklyn, NY

EDUCATION

New York University — May 2026 (expected)
PhD, Urban Systems (GPA: 3.85), Civil and Urban Engineering Department
Relevant Coursework: Machine Learning for Cities, Optimization Methods, Complex Urban Systems, Urban Informatics

University of California, Berkeley — May 2021
Master of Science, Systems Engineering (GPA: 3.96), Civil and Environmental Engineering Department
Relevant Coursework: Data Science, Scalable Spatial Analytics, Behavior Modeling, Energy Systems and Control

University of California, Berkeley — May 2019
Bachelor of Science, Civil and Environmental Engineering (GPA: 3.73)
Relevant Coursework: Computer Programming, Data Analysis, Applied Data Science, Systems Analysis, Cyber-Physical Systems

SKILLS

Coding Languages: Python (NetworkX, OSMnx, scikit-learn, TensorFlow, Geopandas), R
Professional Skills: Bartending (Customer Service, Communication, Problem-Solving, Time Management, Conflict Resolution)

RESEARCH and TEACHING EXPERIENCE

Research Assistant - CUSP, New York University — Aug. 2024 - Present

- Engineered high-dimensional features from large-scale individual mobility data to optimize model performance.
- Built an ML pipeline using XGBoost to classify EV drivers, leveraging mobility patterns to generate data-driven insights.

Research Assistant - Marron Institute, New York University — Jan. 2023 - Present

- Authored literature review analyzing segregation through the lens of mobility patterns and neighborhood dynamics.
- Constructed directed and weighted neighborhood connectivity networks across the entire U.S., for 5 years.
- Analyzed network metrics to determine if neighborhood mobility patterns can create a robust neighborhood typology.

Graduate Student Fellow – Office of Undergraduate Research, New York University — May-June. 2025

- Mentored a cohort of students through Summer Undergraduate Research Incubator (SURI).
- Guided and instructed students in developing an original research proposal over the span of six weeks

Consulting Research Engineer - UC Berkeley — Jan. - Sept. 2022

- Authored proposal and secured funding for a tool to model equitable electrification of zero-emissions mobility.
- Developed metrics to optimize public EV charger placement, considering equity, economic and environmental benefits.
- Led bi-weekly team project meetings, and managed an undergraduate engineering student in conducting research.

Transfer Pre-Engineering Program (T-PREP) Design Assistant- UC Berkeley — July – Aug. 2020, 2021

- Mentored three teams of incoming Berkeley Engineering transfer students from underrepresented backgrounds
- Advised teams through all design stages, including identification of a societal scale problem, cyber-physical system solution design, and pitch development

Associate Data Scientist - Lawrence Berkeley National Laboratory — June - Dec. 2019
Student Research Assistant - Lawrence Berkeley National Laboratory — Jan. 2018 - June 2019

- Built data pipelines using Python to clean real-time occupancy data and optimize building energy use.
- Conducted literature review to identify gaps in Deep Reinforcement Learning control algorithms for building systems.

PROJECTS

Emergency Food Access Index - City Harvest — Jan. 2023 - June 2024

- Developed a multidimensional Emergency Food Access Index to quantify need for resource allocation.
- Simulated travel time using multiple data sources (OsmnX, GTFS, Google Distance-Matrix API, Uber) and ML methods.

- Published paper in *Health & Place*, leading to an invitation from NYC Mayor's Office of Food Policy to discuss findings.

Network Simulation of Police Response Time Under Variable Staffing Levels Fall 2020

- Led development of a Python-based Network Analysis to assess police staffing impact on response time.
- Collaborated with domain experts to develop a robust model, publishing findings in the *European Physics Journal*.

PUBLICATIONS

Clark, Callie, Christa Perfit, and Alice Reznickova. "A multi-dimensional access index: Exploring emergency food assistance in New York City." Health & Place 89 (2024): 103319

Clark, Callie, Chitra Dangwal, Dylan Kato, and Marta Gonzalez. "A network spatial analysis simulating response time to calls for service at variable staffing levels: A case study on strategic police defunding in the city of Chicago." The European Physical Journal Special Topics (2022): 1-9.

FELLOWSHIPS & AWARDS

National Science Foundation Graduate Research Fellow - Highly selective over 13,000 applicants	2020 - 2025
New York University Urban Doctoral Fellow- Competitive Cross-University Program	2023 -2024
ITS Berkeley Statewide Research Transportation Program Grant – Awarded $80,000 Project Budget	2021 – 2022